# Assignment 2

[2025-2] 데이터사이언스를 위한 컴퓨팅의 기초

- Assignment 2 는 세개의 문제로 구성되어있습니다. 주어진 **skeleton.py** 파일을 완성하여 제출해주세요.
- 파일명은 [student_number]_[student_name]으로 하시기 바랍니다.

# Problem 1 (30pt)

### Description

Given two strings, s and t, where t is generated by randomly shuffling string s and then adding one more letter at a random position, the task is to determine the letter that was added to t.

### Constraints

- 0 <= s.length <= 1000
- t.length == s.length + 1
- s and t consist of lowercase English letters.

The test cases are as follows:

| |
|---|
| Input: s = "abcd", t = "abecd"<br>Output: "e"<br>Explanation: 'e' is the letter that was added. |

| |
|---|
| Input: s = "", t = "y"<br>Output: "y" |

# Problem 2 (30pt)

## Description

You have an array called cookies, where each element represents the number of cookies in a bag. Additionally, you're given an integer, k, which signifies the number of children you need to distribute all the bags of cookies to. Each child receives cookies from one or more bags, but the cookies from a single bag cannot be divided among multiple children.

The task is to find the fairest possible distribution, defined as the one where the maximum number of cookies received by any child is minimized.

## Constraints

- 2 <= cookies.length <= 8
- 1 <= cookies[i] <= 105
- 2 <= k <= cookies.length

The test cases are as follows:

Input: cookies = [8,15,10,20,8], k = 2

Output: 31

Explanation: Let's consider one possible distribution: [8, 15, 8] and [10, 20].

The first child receives [8, 15, 8], totaling 8 + 15 + 8 = 31 cookies.

The second child receives [10, 20], totaling 10 + 20 = 30 cookies.

The unfairness of this distribution is the maximum among the totals, which is max(31, 30) = 31.

It's evident that no other distribution yields a fairness lower than 31.

Input: cookies = [6,1,3,2,2,4,1,2], k = 3

Output: 7

Explanation: Let's examine one optimal distribution: [6, 1], [3, 2, 2], and [4, 1, 2].

The first child receives [6, 1], totaling 6 + 1 = 7 cookies.

The second child receives [3, 2, 2], totaling 3 + 2 + 2 = 7 cookies.

The third child receives [4, 1, 2], totaling 4 + 1 + 2 = 7 cookies.

In this distribution, the unfairness is calculated as the maximum among the totals, which is max(7, 7, 7) = 7.

It's evident that no other distribution achieves a fairness lower than 7.

# Problem 3 (40pt)

## Description

Given an undirected tree with nodes labeled from 0 to n - 1 and n - 1 edges represented by a 2Dinteger array edges, where edges[i] = [ai, bi] indicates an edge between nodes ai and bi. Additi onally, you're provided with an integer array called restricted, which lists nodes that are restricted.

Your task is to determine the maximum number of nodes reachable from node 0 without visiting any restricted nodes.

Note that node 0 itself is not restricted.

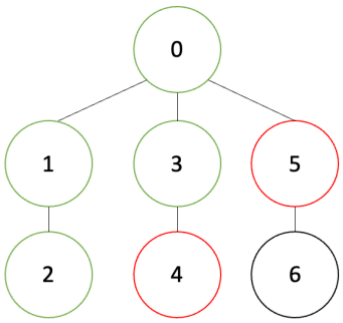
## Constraints

- 2<=n<=105
- edges.length == n – 1
- edges[i].length == 2
- 0<=ai,bi<n
- ai!=bi
- edges represents a valid tree.
- 1 <= restricted.length < n
- 1 <= restricted[i] < n
- All the values of restricted are unique.

The test cases are as follows:

Input: n = 7, edges = [[0,1], [1,2], [3,0], [4,3], [0,5], [5,6]], restricted = [4,5]

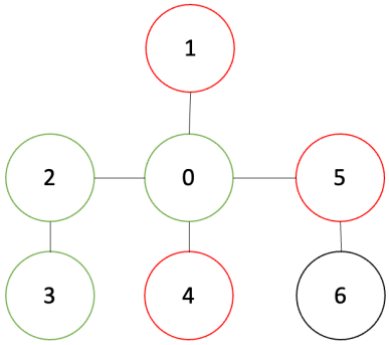
Output: 4




Explanation: In the illustrated tree diagram, nodes [0, 1, 2, 3] are the only ones accessible from node 0 without encountering any restricted nodes.

Input: n = 7, edges = [[0,1], [0,2], [0,5], [0,4], [3,2], [6,5]], restricted = [4,5,1]

Output: 3




Explanation: Illustrated in the tree diagram above, nodes[0,2,3]arethesoleaccessiblenodesfromnode 0 without encountering any restricted nodes.